

ERD Design

The MoMo SMS Processing System's database form was made to help improve the efficiency of processing, keeping, and overview of mobile money sms transactions. The method makes sure the structure is complex enough for a startup-level implementation while minimizing redundancy and enforcing data consistency by following the normalization obligations.

The names, phone numbers, and emails of the clients are being kept in the USERS table. There is a quite simple relationship between USERS and transactions, which is one-to-many since a single user can do a lot of transactions without any sort of limit. This website will be able to keep and fetch original SMS data for auditing and debugging due to this relationship.

The money, transaction code, recipient details, and balances are all kept in a wonderfully structured transaction details in the TRANSACTIONS table. We have connected all tables in an efficient way so as to facilitate tracing data. Transaction categories are categorized and put into a TRANSACTION_TYPES table to make sure scarcity and preserve clean data organization. This really helps the rise of transaction categories and avoids repetition of transaction type names across records.

Account management is quite simple through the ACCOUNTS table, which we use to track a client's balance and account info separately from transaction records.

This ERD design was chosen because it is modular, scalable, and easy to maintain. It supports future improvements such as fraud detection, APIs, notifications, and advanced analytics while remaining simple enough for an academic project and early-stage system deployment.

We chose this ERD diagram because it is simple to use and efficient, and it will help in future improvements as we can add and modify the tables without affecting its core functions.